Exercise 6 • Full Accounting Cycle & Uncalled Capital

Company X Full Accounting Cycle, Balance Sheet & Uncalled Capital

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 3

Company X incorporated on 1/1/2017 with 100 m.u. shares fully paid in cash. Purchased on 1/1/2017 with cash: a) land/area for 20 m.u., b) equipment with life of 10 years (straight-line) for 20 m.u., c) raw materials for 20 m.u.
Hires an employee for annual cost of 10 m.u. Employee processes half of raw material and sells it for 50 m.u. Sales are uniform every day, average collection period is 3 months. Corporate tax is 30% (paid on 31/12/2017). Strict self-financing strategy.
***1.** Prepare the Balance Sheet as on 31/12/2017 with intermediate calculations.*
***2.** Indicate and explain changes if shareholders had paid only 90% of the 100 m.u. subscribed.*

EXAM DAY STRATEGY: FULL ANNUAL CYCLE, TREASURY LEDGER & CONTRA-EQUITY

- **Opening Cash Position on 1/1/2017:**

$$\text{Cash}_{1/1} = \text{Share Capital} - (\text{Land} + \text{Equipment} + \text{Materials}) = 100.0 - (20.0 + 20.0 + 20.0) = \mathbf{40.0}\ \text{mu}$$

- **Operating Cycle & COGS:** Only half of raw materials are processed $\implies \text{COGS} = 20.0/2 = \mathbf{10.0}$ mu. The remaining 10.0 mu remains in warehouse inventory as a **Current Asset**.
- **Exam Trap 1 – Depreciation of Land: Land is indestructible and never depreciates!** Only Equipment depreciates ($20.0/10 = \mathbf{2.0}$ mu). Depreciating land is an immediate exam penalty.
- **Exam Trap 2 – Accounts Receivable Delay:** Sales occur uniformly over 365 days with an average collection period of 3 months (0.25 year).

$$\text{Ending Receivables} = 50.0 \times \frac{3\ \text{months}}{12\ \text{months}} = \mathbf{12.5}\ \text{mu} \implies \text{Collections} = 50.0 - 12.5 = \mathbf{37.5}\ \text{mu}$$

  Never assume that all 50 mu of sales were collected in cash!
- **Exam Trap 3 – Corporate Tax Settlement:** The problem states that corporate tax ($30\% \times 28 = 8.4$ mu) is *paid on December 31st, 2017*. Thus, it is a **cash outflow** during 2017, leaving **zero tax liability** pending on the year-end balance sheet!
- **Exam Trap 4 – Question 2 (Uncalled Capital):** If shareholders only pay 90%, uncalled capital is 10 mu. Uncalled capital does NOT change revenues, expenses, or taxes. It decreases starting cash by 10 mu (ending cash becomes 49.1 mu) and is presented in Equity as a negative contra-equity line ($100 - 10 + 19.6 = 109.6$ mu).

Phase 1: Operational Cycle & P&L Statement (Question 1)

1. **Sales Revenue:** Sales $=$ **50.0** m.u.
2. **Raw Materials Consumed (COGS):**

$$\text{COGS} = \frac{20.0}{2} = \mathbf{10.0}\ \text{m.u.} \quad (\text{Ending Inventory stored in warehouse} = 10.0\ \text{m.u.})$$

$$\text{Gross Profit} = 50.0 - 10.0 = \mathbf{40.0}\ \text{m.u.}$$

3. **Operating Expenses (Gastos de Explotación):**
   - Employee labor cost: 10.0 m.u.
   - Equipment depreciation (20/10 yrs): 2.0 m.u.

$$\text{Total Operating Expenses} = 10.0 + 2.0 = \mathbf{12.0}\ \text{m.u.}$$

4. **Operating Income (EBIT) & EBT:**

$$\text{EBIT} = 40.0 - 12.0 = \mathbf{28.0}\ \text{m.u.} \implies \text{EBT} = \mathbf{28.0}\ \text{m.u.}$$

5. **Corporate Income Tax (30% paid in cash on 31/12):**

$$\text{Corporate Tax} = 30\% \times 28.0 = \mathbf{8.4}\ \text{m.u.}$$

6. **Net Income & Self-Financing Appropriation:**

$$\text{Net Income} = 28.0 - 8.4 = \mathbf{19.6}\ \text{m.u.}$$

Under strict self-financing, $\text{Dividends} = 0$ m.u. and the full profit is retained into reserves:

$$\text{Retained Earnings (Addition to Equity)} = +\mathbf{19.6}\ \text{m.u.}$$

Phase 2: Working Capital & Treasury Cash Reconciliation

1. **Accounts Receivable (Trade Debtors):**

$$\text{Ending Accounts Receivable} = \text{Sales} \times \frac{3 \text{ months}}{12 \text{ months}} = 50.0 \times \frac{1}{4} = \mathbf{12.5} \text{ m.u.}$$

$$\text{Cash Collections from Customers} = 50.0 - 12.5 = \mathbf{37.5} \text{ m.u.}$$

2. **Treasury Cash Ledger Reconciliation on 31/12/2017:**

Cash Ledger Item	Amount (m.u.)
Cash balance on 1/1/2017 (100 equity − 60 capex & stock)	40.0
(+) Collections from customers	+37.5
(-) Employee salaries paid	(10.0)
(-) Corporate tax paid on 31/12	(8.4)
Ending Cash Balance on 31/12/2017	**59.1**

Phase 3: Balance Sheet as on 31/12/2017 (100% Paid — Question 1)

ASSETS (Activo)	Amount (m.u.)	LIABILITIES & EQUITY (Pasivo y PN)	Amount (m.u.)
NON-CURRENT ASSETS (Activo No Corriente)	**38.0**	**STOCKHOLDERS' EQUITY (Patrimonio Neto)**	**119.6**
• Land / Industrial Area (Non-depreciable)	20.0	• Subscribed & Paid-in Share Capital	100.0
• Productive Equipment (Historical cost)	20.0	• Retained Earnings (Year 2017 Net Profit)	19.6
• (-) Accumulated Depreciation: Equipment	(2.0)		
CURRENT ASSETS (Activo Corriente)	**81.6**	**NON-CURRENT LIABILITIES**	**0.0**
• Inventories: Unprocessed Raw Materials	10.0	• Long-term bank borrowing	0.0
• Accounts Receivable (Customers, 3-month term)	12.5	**CURRENT LIABILITIES**	**0.0**
• Cash & Cash Equivalents (Treasury account)	59.1	• Accounts payable / Unpaid taxes	0.0
TOTAL ASSETS	**119.6**	**TOTAL LIABILITIES & EQUITY**	**119.6**

Phase 4: Impact of Uncalled Capital (90% Paid — Question 2)

If shareholders contributed only 90% of subscribed shares in cash (90 m.u.), the remaining 10 m.u. represents **Uncalled Capital** (*Accionistas por desembolsos no exigidos*):

1. **Income Statement & Net Profit:** Completely unaffected (Net Income $=$ **19.6** m.u.). Operations, billing, wages, depreciation, and taxes do not depend on the equity disbursement fraction.
2. **Cash & Current Assets:** Starting cash on 1/1/2017 was $90.0 - 60.0 = 30.0$ m.u. (exactly 10.0 m.u. lower).

$$\text{New Ending Cash} = 59.1 - 10.0 = \mathbf{49.1} \text{ m.u.}$$

$$\text{New Current Assets} = 10.0 + 12.5 + 49.1 = \mathbf{71.6} \text{ m.u.}$$

$$\text{New Total Assets} = 38.0 \text{ (NCA)} + 71.6 \text{ (CA)} = \mathbf{109.6} \text{ m.u.}$$

3. **Stockholders' Equity (Statutory Contra-Equity Presentation):** Under standard European / Spanish General Accounting Plan rules, uncalled capital appears as a negative contra-equity line directly beneath subscribed capital:

Equity Component (Question 2)	Amount (m.u.)
Subscribed Share Capital	100.0
(-) Shareholders for Uncalled Capital	(10.0)
(+) Retained Earnings (Year 2017)	+19.6
Total Stockholders' Equity	**109.6**

BALANCE CHECK WITH UNCALLED CAPITAL

$$\text{Total Assets } (\mathbf{109.6} \text{ m.u.}) \equiv \text{Total Liabilities \& Equity } (\mathbf{109.6} \text{ m.u.})$$

Alternative Accounting Presentation Note: Under US GAAP, uncalled capital is occasionally presented as a receivable asset. However, under standard Spanish PGC and IAS/IFRS, presenting it as a negative contra-equity line within Net Equity is the standard, ensuring that net equity accurately reflects resources physically contributed plus retained profit.

Study notes written from the official course material (T3_Financial Statements_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-financial-statements.html